



Name:

Teacher:

SCEGGS Darlinghurst

HSC Assessment 1
Friday, 24th March, 2006

Extension 1 Mathematics

General Instructions

- Time allowed – 75 minutes
- Weighting 25%
- This paper has **four** questions
- Attempt **all** questions and show all necessary working
- Marks may be deducted for careless or badly arranged work
- Write using blue or black pen, diagrams in pencil
- Write your name and your teacher's name at the top of each page
- Approved calculators, mathematical templates and geometrical instruments may be used
- A table of standard integrals is provided at the back of this paper

Questions	Total	Comm.	Reas.	Calc.
1	/10		/3	/4
2	/12	/3	/7	
3	/10	/3	/7	
4	/11	/2	/6	/3
TOTAL	/43	/8	/23	/7

Question 1 (10 marks)

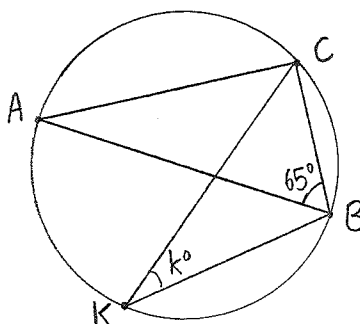
Marks

(a) Find: $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x}$

1

(b)

3

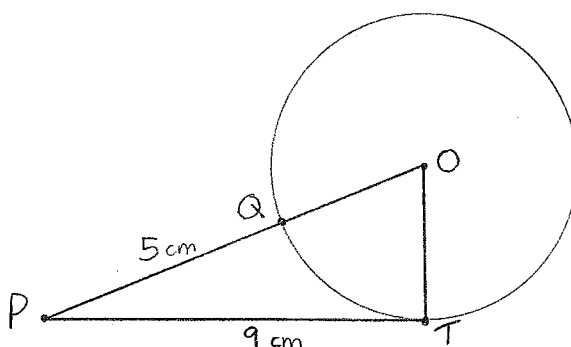


In the diagram above, AB is the diameter of a circle, $\angle ABC = 65^\circ$, and $\angle BKC = k^\circ$.

Find the value of k , giving full reasons for your answer.

(c)

2



In the diagram above, O is the centre of a circle with PT the tangent at T . If $PT = 9$ cm and $PQ = 5$ cm, calculate the length of the radius of the circle. Give full reasons for your answer.

(d) i. Prove that the equation

4

$$2 \sin x - 10x + 5 = 0$$

has a solution between $x = 0$ and $x = 1$.

ii. Taking $x = 0.5$ as a first approximation, find a better approximation with one application of Newton's method (correct to 3 significant figures).

Question 2 begins on page 2 ...

START A NEW PAGE

Question 2 (12 marks)

Marks

- (a) By making the substitution $t = \tan(\frac{\theta}{2})$, or otherwise, show that 2

$$\cot\theta + \tan\frac{\theta}{2} = \operatorname{cosec}\theta$$

- (b) i. Express $\sqrt{3}\sin t + \cos t$ in the form $R\sin(t + \alpha)$, where α is in radians. 2

- ii. Hence, or otherwise, find (in exact form) the general solution of the equation 2

$$\sqrt{3}\sin t + \cos t = 1$$

- (c) Use mathematical induction to prove that, for any positive integer n , $5^n + 2(11)^n$ is a multiple of 3. 3

- (d) An employer wishes to choose two people for a job. There are 8 applicants, 3 of whom are women and 5 of whom are men. 3

- i. If each applicant is interviewed separately and all the women are interviewed before any of the men, find how many ways there are to carry out the interviews.
- ii. How many ways can two applicants be chosen so that at least one of those chosen is a woman.

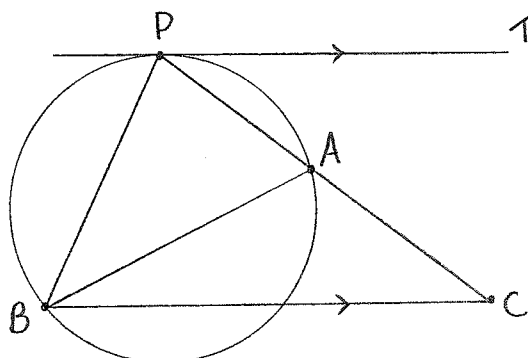
Question 3 begins on page 3 ...

START A NEW PAGE

Question 3 (10 marks)

Marks

(a)



In the diagram, A , P , and B , are points on the circle. The line PT is tangent to the circle at the point P , and PA is produced to C so that BC is parallel to PT .

- i. Show that $\angle PBA = \angle PCB$ 2
- ii. Deduce that $PB^2 = PA \times PC$ 2

(b) 5 males and 5 females are seated around a circular table. 3

- i. How many seating arrangements are possible if there are no restrictions?
- ii. If no two people of the same sex are to sit next to each other, how many arrangements are possible?
- iii. If Amy refuses to sit next to Benton, and there are no other restrictions, how many seating arrangements are possible?

(c) Use mathematical induction to prove that, for all positive integers n , 3

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

Question 4 begins on page 4 ...

START A NEW PAGE

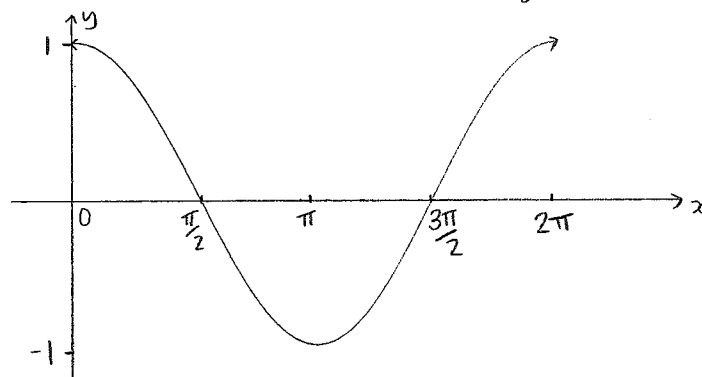
Question 4 (11 marks)

Marks

(a) In the following questions you may use the identity

$$\cos^2 x = \frac{1}{2} \cos 2x + \frac{1}{2}.$$

- i. Copy the sketch of $y = \cos x$ (below) into your writing booklet, and on the same set of axes sketch the curve $y = \cos^2 x$. 2



- ii. Show that $\cos^4 x = \frac{3}{8} + \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x$ 2
- iii. Hence, find the volume generated when the area bounded by the curves $y = \cos x$ and $y = \cos^2 x$, between $x = 0$ and $x = \frac{\pi}{2}$, is rotated about the x -axis. 3

(b) Let each different arrangement of all the letters of PERMUTATION be called a word.

- i. How many words are possible? 1
- ii. If a word is selected at random, what is the probability that all the vowels are together? 2
- iii. How many words are there in which the vowels occur in the order AEIOU from left to right, though not necessarily together? 1

END OF ASSESSMENT

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

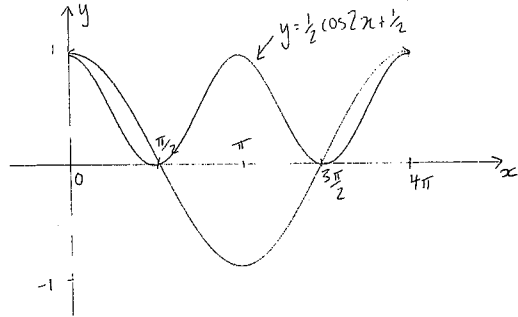
Question 4 (11 marks)

(a)(i) $y = \cos^2 x$

$$= \frac{1}{2} \cos 2x + \frac{1}{2}$$

stretch squash shift up.

✓ 2 correct transformations
✓ all correct.



Comm 2.

(ii) LHS = $\cos^4 x$

$$\begin{aligned} &= (\cos^2 x)^2 \\ &= \left(\frac{1}{2} \cos 2x + \frac{1}{2}\right)^2 \\ &= \frac{1}{4} \cos^2 2x + \frac{1}{2} \cos 2x + \frac{1}{4} \\ &= \frac{1}{4} \left(\frac{1}{2} \cos 4x + \frac{1}{2}\right) + \frac{1}{2} \cos 2x + \frac{1}{4} \\ &= \frac{1}{8} \cos 4x + \frac{1}{8} + \frac{1}{2} \cos 2x + \frac{1}{4} \\ &= \frac{3}{8} + \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x \end{aligned}$$

Reas 2.

$$* (A+B)^2 = A^2 + \underline{2AB} + B^2$$

* If they give you some information then USE IT!

(iii) $V = \pi \int_0^{\pi/2} y_{\text{top}}^2 - y_{\text{bottom}}^2 dx$

$$= \pi \int_0^{\pi/2} \cos^2 x - \cos^4 x dx$$

$$= \pi \int_0^{\pi/2} \frac{1}{2} \cos 2x + \frac{1}{2} - \left(\frac{3}{8} + \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x\right) dx$$

$$= \pi \int_0^{\pi/2} \frac{1}{8} - \frac{1}{8} \cos 4x dx$$

$$= \frac{\pi}{8} \int_0^{\pi/2} 1 - \cos 4x dx$$

$$* V = \pi \int \text{top}^2 - \text{bottom}^2$$

NOT $\pi \int (\text{top} - \text{bottom})^2$
(this got you into a mess)

* use the previous parts of the question.

$$V = \frac{\pi}{8} \int_0^{\pi/2} 1 - \cos 4x dx$$

$$= \frac{\pi}{8} \left[x - \frac{\sin 4x}{4} \right]_0^{\pi/2} \checkmark$$

$$= \frac{\pi}{8} \left[\left(\frac{\pi}{2} - \frac{\sin 2\pi}{4} \right) - \left(0 - \frac{\sin 0}{4} \right) \right]$$

$$= \frac{\pi^2}{16} \text{ cube units } \checkmark$$

Calc. 3

(b) (i) PERMUTATION

$$\# \text{ words} = \frac{11!}{2!} \checkmark$$

Reas. 4

(ii) Prob (all vowels together)

$$= \frac{\# \text{ ways vowels all together}}{\text{total \# ways}} \checkmark$$

$$= \frac{7! \times 5!}{11!} \checkmark$$

All vowels together:

E, U, A, I, O
P R M T T N

7! ways to arrange group (2 Ts)
x 5! ways to arrange vowels

$$= \frac{7! \times 5!}{11!}$$

* always be on the lookout for repeated letters!

* vowels can be rearranged (they are not simply written on a card 'AEIOU')

* Remember to read the question carefully to see if they want a probability or not.

(iii) -----

11 positions

→ $\binom{11}{5}$ ways to choose 5 positions for the vowels A, E, I, O, U,
& only 1 way to arrange those vowels.

→ $\frac{6!}{2!}$ ways to then arrange consonants.

$$\therefore \# \text{ words} = \binom{11}{5} \times \frac{6!}{2!} \checkmark$$

* Hard question! well done to those who solved it.

* Alternative solution:
 $\frac{1}{5!}$ of the total arrangements will have the vowels in order

$$\therefore \# \text{ words} = \frac{1}{5!} \times \frac{11!}{2!}$$

Question 3 (10 marks)

(a)(i) $\angle PBA = \angle TPA$ ($\angle$ bet. tang. & chord
= $\angle$ in alt. seg.) ✓

$\angle TPA = \angle PCB$ (alt. $\angle$ s $\parallel$ =) ✓

$\therefore \angle PBA = \angle PCB$

(ii) In $\triangle PBA$ & $\triangle PCB$

$\angle PBA = \angle PCB$ (part i)

$\angle BPA = \angle CPB$ (common)

$\therefore \triangle PBA \sim \triangle PCB$ (equiangular)

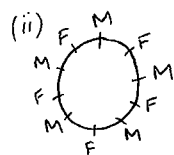
$\therefore \frac{PA}{PB} = \frac{PB}{PC}$ (corr sides in similar
 $\triangle$ s in same ratio)

$PB^2 = PA \times PC$ ✓

(b) (i) # unrestricted
seating
arrangements

= $1 \times 9!$
way to
seat 1st
person ways to
arrange
rest

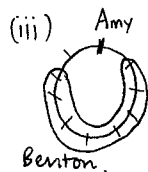
= $9!$
= 362 880 ✓



ways = $1 \times 4! \times 5!$

way to seat
1st (say, M) ways to
seat
males ways to
seat
females

= $4! 5!$
= 2880 ✓



ways = $1 \times 7 \times 8!$

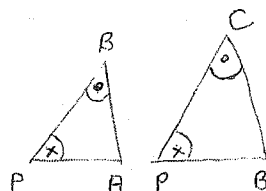
ways to
seat 1st
(Amy)
say. ways
to seat
Benton ways
to seat
rest

= $7 \times 8!$ ✓

OR, # ways = total
ways - # ways A, B
together

= $9! - 1 \times 8! \times 2!$
= 282240 ✓

Reas. 4



Note the order of the letters
on the similar $\triangle$ s
put them/draw them in
the same orientation

Reas 3

This question
was very well
done.

(c) Prove $1 \times 1! + 2 \times 2! + \dots + n \times n! = (n+1)! - 1$

* Prove true for $n=1$

LHS = $1 \times 1! = 1$

RHS = $(1+1)! - 1 = 1$ $\therefore$ true for $n=1$ ✓

* Assume true for $n=k$

ie. assume $1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$

* Prove true for $n=k+1$

ie. prove $1 \times 1! + \dots + k \times k! + (k+1)(k+1)! = (k+2)! - 1$

LHS = $(k+1)! - 1 + (k+1)(k+1)! \checkmark$
using
assumption

= $(k+1)! + (k+1)(k+1)! - 1$

= $(k+1)! (1 + k+1) - 1$

= $(k+1)! (k+2) - 1$

= $(k+2)! - 1$

= RHS ✓

$\therefore$ If true for $n=k$, it's true for $n=k+1$,
since true for $n=1$, it's true for $n=2, 3, \dots$

$\therefore$ By PMI, it's true for all integers $n > 0$.

Comm 3

Please set working out
as two separate parts

LHS =

RHS =

$\therefore$ LHS = RHS.

Rearrange if necessary
and factorize out
 $(k+1)!$

No fudging please.
Yes! It's obvious if you do!

Question 2 (12 marks)

(a) $\cot \theta + \tan \theta/2$

$$= \frac{1}{\tan \theta} + \tan \theta/2$$

$$= \frac{1-t^2}{2t} + t \quad \checkmark \quad \left[\text{since } \tan \theta = \frac{2t}{1-t^2} \right]$$

$$t = \tan \theta/2$$

$$= \frac{1-t^2+2t^2}{2t}$$

$$= \frac{1+t^2}{2t}$$

$$= \frac{1}{\sin \theta} \quad \checkmark \quad \left[\text{since } \sin \theta = \frac{2t}{1+t^2} \right]$$

$$= \operatorname{cosec} \theta$$

(b) (i) $\sqrt{3} \sin t + \cos t$

$$R \sin t \cos \alpha + R \cos t \sin \alpha = R \sin(t + \alpha)$$

$$R \cos \alpha = \sqrt{3} \quad \textcircled{1}$$

$$R \sin \alpha = 1 \quad \textcircled{2}$$

$$\textcircled{2} : \tan \alpha = 1/\sqrt{3} \quad \textcircled{1}^2 + \textcircled{2}^2 : R^2 = \sqrt{3}^2 + 1^2$$

$$\textcircled{1} \quad \alpha = \pi/6 \quad \checkmark \quad R = 2 \quad \checkmark$$

$$\therefore \sqrt{3} \sin t + \cos t = 2 \sin(t + \pi/6)$$

(ii) $\sqrt{3} \sin t + \cos t = 1$

$$2 \sin(t + \pi/6) = 1$$

$$\sin(t + \pi/6) = 1/2$$

$$(t + \pi/6) = \pi n + (-1)^n (\sin^{-1} 1/2)$$

$$t + \pi/6 = \pi n + (-1)^n \pi/6$$

$$t = \pi n + (-1)^n \pi/6 - \pi/6 \quad \checkmark \checkmark$$

[✓ for solutions between 0 & 2π]

This question was very well done.

Reas. 4

Those that used the formula did much better - although you have to be careful to use the formula correctly at the right spot.

(c) Prove $5^n + 2(11)^n$ is a multiple of 3.

* Prove true for $n=1$

$$5^1 + 2(11)^1 = 27 = 3 \times 9 \quad \therefore \text{true for } n=1 \quad \checkmark$$

* Assume true for $n=k$

$$\text{ie. assume } 5^k + 2(11)^k = 3M \quad [M \text{ integer}]$$

$$5^k = 3M - 2(11)^k$$

* Prove true for $n=k+1$

$$\text{ie. prove } 5^{k+1} + 2(11)^{k+1} = 3 \times \text{stuff}$$

$$5^{k+1} + 2(11)^{k+1}$$

$$= 5 \times 5^k + 2 \times 11 \times 11^k$$

$$= 5(3M - 2 \times 11^k) + 22 \times 11^k \quad \checkmark$$

$$= 15M - 10 \times 11^k + 22 \times 11^k$$

$$= 15M + 12 \times 11^k$$

$$= 3(5M + 4 \times 11^k) \text{ which is a multiple of 3} \quad \checkmark$$

* Therefore, if it's true for $n=k$, it's true for $n=k+1$. Since it's true for $n=1$, it's true for $n=2, 3, \dots$

$$\therefore \text{by PMI it's true for all integers } n > 0.$$

comm. 3

* Please learn your index rules - never ever multiply bases!

* And don't substitute twice ($5^k = 3M - 2(11)^k$ & $2(11)^k = 3M - 5^k$)

It just won't fall out if you do.

(d) $W_1 W_2 W_3 M_1 M_2 M_3 M_4 M_5$

(i) $3! \times 5!$ ways to carry out interviews $\checkmark$

↑
arrange women 1st
↑
arrange men next.

(ii) # ways to choose 2, = total # ways choose 2 - # ways to choose so no women $\checkmark$

$$= \binom{8}{2} - \binom{5}{2}$$

$$= 18$$

OR # ways ≥ 1 W = # ways W, M + # ways WW $\checkmark$

$$= \binom{3}{1} \binom{5}{1} + \binom{3}{2} = 18 \quad \checkmark$$

Reas. 3

* AND $\Rightarrow$ MULTIPLY
* OR $\Rightarrow$ ADD

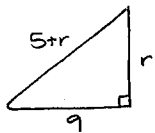
In general, this question was done well.

Question 1 (10 marks)

(a) $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{3}{2} \cdot \frac{2x}{\sin 2x} = \frac{3}{2} \checkmark$

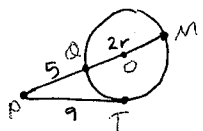
(b) $\angle ACB = 90^\circ$ ($\angle$ in a semicircle $= 90^\circ$) $\checkmark$
 $\angle CAB = 25^\circ$ ($\angle$ sum $\triangle ABC = 180^\circ$) $\checkmark$
 $\angle CKB = \angle CAB = 25^\circ$ ($\angle$ s in same seg $=$) $\checkmark$
 $\therefore k^\circ = 25^\circ$

(c) $\angle PTO = 90^\circ$ (tangent $\perp$ radius at pt of tang) $\checkmark$



Pythagoras: $\rightarrow$
 $(5+r)^2 = r^2 + 9^2$
 $25 + 10r + r^2 = r^2 + 81$
 $10r = 56$
 $r = 5.6 \text{ cm} \checkmark$

OR



Product of the intercepts on the intersecting tangent & secant are equal. $\checkmark$
 $5(5+2r) = 9^2$
 $25 + 10r = 81$
 $10r = 56$
 $r = 5.6 \text{ cm} \checkmark$

(d) (i) $f(x) = 2\sin x - 10x + 5 = 0$

$f(0) = 2\sin 0 - 10 \times 0 + 5 = 5 > 0$

$f(1) = 2\sin 1 - 10 \times 1 + 5 = -3.3 < 0 \checkmark$

& $f(x)$ is continuous. $\checkmark$

$\therefore 2\sin x - 10x + 5 = 0$ has a solution $0 < x < 1$

(ii) $f'(x) = 2\cos x - 10$

$x_1 = 0.5$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \checkmark$

$= 0.5 - \frac{(2\sin 0.5 - 10 \times 0.5 + 5)}{(2\cos 0.5 - 10)}$

$= 0.616297... (2\cos 0.5 - 10)$

$= 0.616$ to 3 sig. fig. $\checkmark$

Show working steps.

Reas 3. Each reason required for the mark.
 Learn the correct wording of the reasons.

This method is more successful except when algebraic errors occur in expanding $(5+r)^2$

point must be on circumference

Note

$PT^2 = PQ \times PM$

not

$PT^2 = PQ \times QO$

or

$PT^2 = PQ \times PO$

Calc. 4

must state curve is continuous. This word is very important.

Is your calculator in RADIAN mode?